

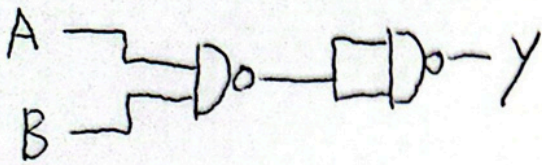
4-Bit Accumulator

E V A N
C H E N

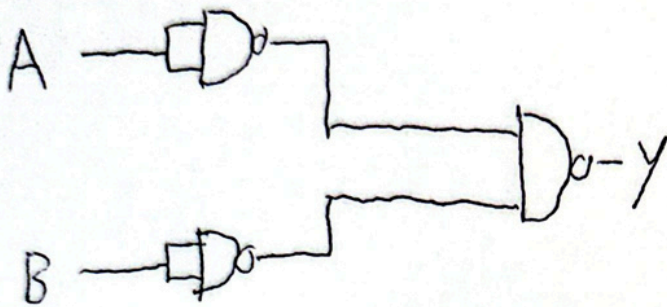
2026

ROBOTICS/ELECTRICAL ENGINEERING

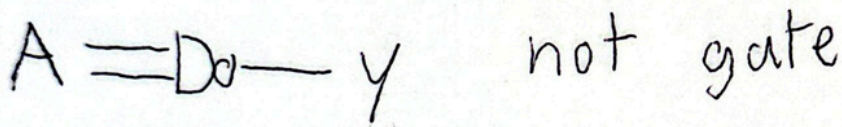
Power of Nand



and gate



or gate

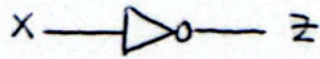


not gate

Truth tables

74HC04 (Not) $x = \bar{x}$

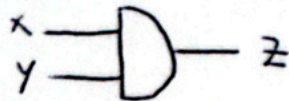
X	Z
0	1
1	0



74HC08 (And)

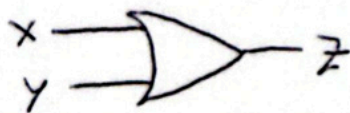
x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

$$x \cdot y = z$$



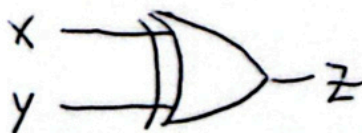
74HC32 (Or) $x + y = z$

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1



74HC86 (Xor) ~~z~~ $x\bar{y} + \bar{x}y = z$

x	y	z
0	0	0
0	1	1
1	0	1
1	1	0



CPU 74HC

Logic gates

74HC00

74HC04

74HC08

74HC32

74HC86

Decoder

74HC138

Arithmetic

74HC283

Registers

74HC173

74HC273

Bus

74HC245

74HC244

Mux

74HC157

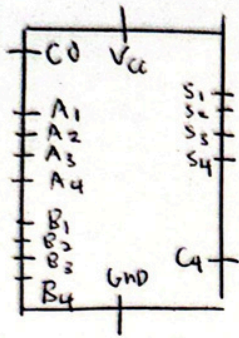
Counters

74HC163

Editing using symbol editor

Original

name: 74LS 283



4-bit full adder

74HC173 control inputs

symbol/name	connection	why
O _{e1}	0 GND	} enables output
O _{e2}	GND	
E ₁	GND	} enables input
E ₂	GND	
Cp	Manual clock	updates on clock edge
Mr	+5v	prevents reset

EC-4 CPU

CPU must process 4 bits at a time

bit = 0 or 1

4 bit value = $2^4 = 16$ possible digits

0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

4 bit data bus

	data 0	data 1	data 2	data 3
example data 10	1	0	1	0

* each wire carries one bit

* bigger cpu =
bigger PCB

Program Counter

- main purpose is to understand which instructions to execute next

Line 0: LDI 5
Line 1: ADD 3
Line 2: OUT
Line 3: HLT

- * program counter will be a 4 bit register

PC3	PC2	PC1	PC0
0	1	1	0

means address 6

shopping list

- 74HC04 (not gate)
- 74HC08 (And gate)
- 74HC32 (or gate)
- 74HC86 (Xor gate)

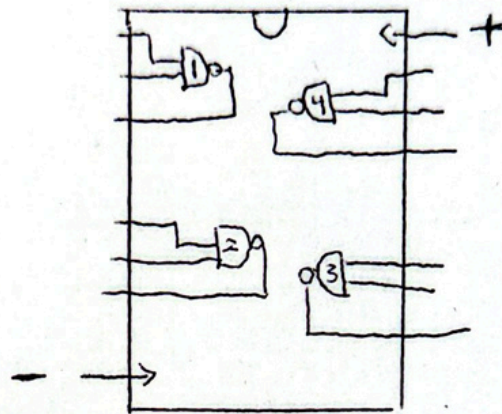
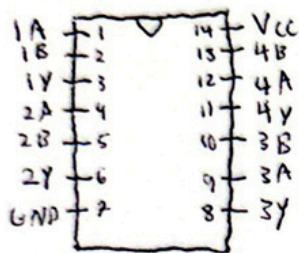
SN 74C00N

SN = TI prefix

74HC = High speed CMOS

00 = Quad 2input Nand

N = Dip package



Operating voltage: 2V-6V

Nand gate 1

1A	1B	1Y
-	-	+
-	+	+
+	-	+
+	+	-

Nand gate 2

2A	2B	2Y
-	-	+
-	+	+
+	-	+
+	+	-

Nand gate 3

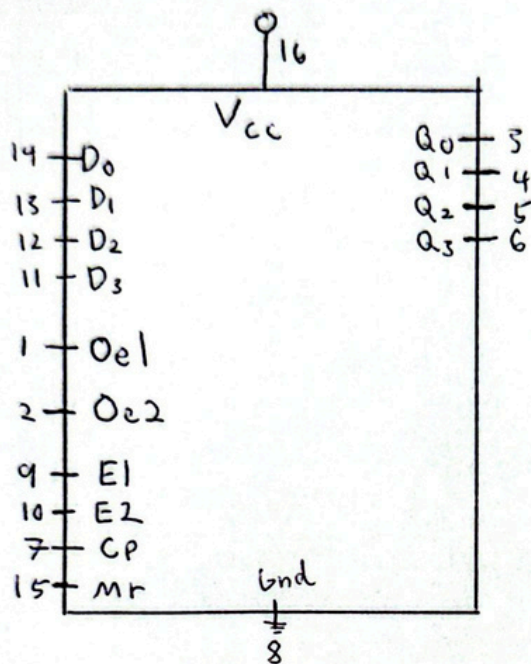
3A	3B	3Y
-	-	+
-	+	+
+	-	+
+	+	-

Nand gate 4

4A	4B	4Y
-	-	+
-	+	+
+	-	+
+	+	-

Accumulator

74HC173



- 4 flip-flops slammed together

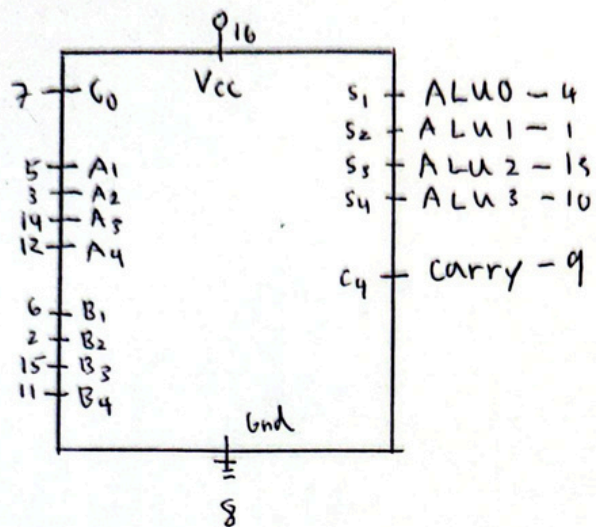
<u>inputs</u>	<u>outputs</u>
D ₀	Q ₀
D ₁	Q ₁
D ₂	Q ₂
D ₃	Q ₃

example: $1011 \xrightarrow{D_n} \text{input} \xrightarrow{\text{clock edge up}} \text{output} \xrightarrow{Q_n} 1011$

- Oe₁ and Oe₂ must both be active for outputs to be enabled
- Cp is clock pulse
- Mr is master reset
- E₁ and E₂ must both be active for inputs to be enabled

Arithmetic Logic Unit

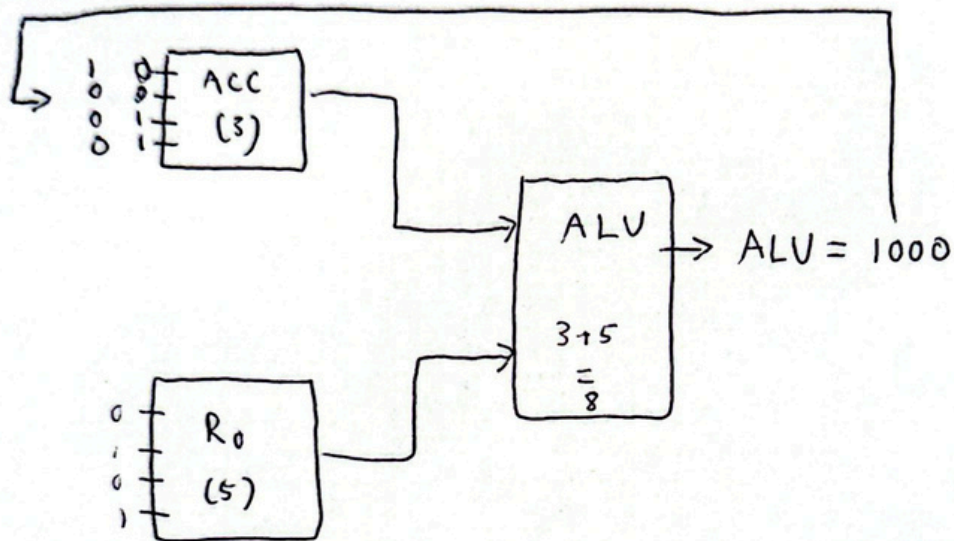
74HC283



ALU back to ACC

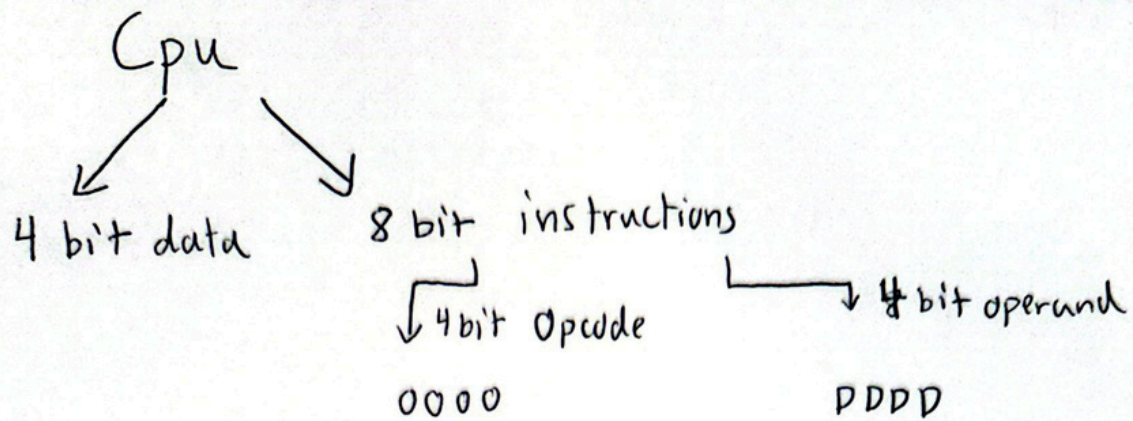
Suppose

$\text{Acc} = 0011^{(3)}$ and $R_0 = 0101^{(5)}$



- our initial value of $\text{Acc} = 0011$ is replaced by 1000 when clock edge rises

- Acc's new value (8) would then be added to (5) in the ALU to get a new value that's put into the Acc again



* values calculated are in 4 bits

8 bit instructions are divided into 2 segments:

4 bit opcode
- describe operation

4 bit operand
- address / where

example

definition

0001 = LDI

0101 = 5

∴ ~~0001 0101~~ 0001 0101 means
load the number 5

* the instruction itself is 8 bits long
but we can only load a value of 4 bits
(0-15)

Button

